

TECHNICAL SPECIFICATION

DOCUMENT 00

Ultra-Safe Central Capsule

Scalable Survival Module for Ground, Aerial & Maritime Platforms

Document Version	Date	Status
1.3 Full Detailed (6 Independent Safety Layers)	August 2026	Complete Baseline for Design, Simulation, Testing & Patent Extraction

This document establishes the complete and authoritative technical baseline for the Ultra-Safe Central Capsule. It defines geometry, mass properties, structural architecture, the six independent safety layers, parachute and host-separation systems, impact energy management, life-support and environmental systems, flotation and self-righting capability, emergency location, fire protection, host-platform interfaces, verification strategy, and preliminary patent-family assessment. The document is intended to serve as the single reference for all subsequent detailed design, analysis, simulation, testing and intellectual-property activities.

Revision Note (Version 1.3 Full Detailed): The safety architecture has been expanded from four to six fully independent layers. New content includes a detailed mass budget, flotation and self-righting systems, dual-band emergency location, refined structural load-path description, host-interface requirements, a verification and validation outline, and an updated patent-family assessment. Door configuration is retained as two side doors plus one large reinforced roof hatch, with a fully continuous floor plane. Occupant capacity remains seven persons arranged in a circular/radial seating layout. This version is substantially expanded to provide the depth required for detailed engineering and patent extraction.

1. Purpose and Design Intent

The Ultra-Safe Central Capsule is an independent, modular survival cell designed for installation at the structural heart of a wide range of host platforms. These include ground vehicles, helicopters, fixed-wing aircraft, aerial work platforms and cranes, and large maritime structures such as the Fast Ocean Pyramid family. The single overriding requirement is that the occupants remain alive and recoverable after the complete loss of the host vehicle, whether that loss is caused by high-energy impact, uncontrolled fire, explosion, structural break-up, or submersion.

The capsule is not a conventional escape pod that relies on a single recovery method. Instead it implements a multi-layer, multi-domain survival architecture that remains functional even when several critical systems have failed. After separation the capsule is completely independent of the host for power, control, life support, location and recovery. This independence is the foundation of the entire safety philosophy.

1.1 Core Design Principles

1. **Functional Independence** – After separation the capsule requires no services from the host vehicle. All power, control, life support and location functions are self-contained.
2. **Six Independent Safety Layers** – Every critical survival function is protected by six independent paths. No single-point failure, whether electrical, mechanical, pyrotechnic or structural, can disable the overall survival sequence.
3. **Structural Continuity** – Geometry and structure are scalable while preserving a continuous high-strength roll-cage and an unbroken floor plane that maximises strength for the landing system.
4. **Multi-Axis Integrity** – The structure and energy-absorption system are designed for extreme combined loads including vertical impact, lateral impact, roll-over and full inversion.

5. **End-to-End Survival Sequence** – The design addresses the complete chain: detection of the emergency, clean separation from the host, controlled descent, survivable landing, and long-duration post-landing support and location.

6. **Certification Orientation** – Architecture, documentation and verification planning are structured to support both aviation and maritime certification pathways.

2. Geometry, Occupancy and Mass Properties

The standard-size capsule is defined by a cylindrical external envelope of 3.60 m diameter and 2.80 m overall height (including the stowed landing-leg envelope). Internal usable volume after structure and systems is approximately 18–20 m³. Seven occupants are accommodated in a circular/radial seating arrangement that places the pilot and co-pilot positions for optimal access to controls while distributing G-loads more evenly among all occupants and preserving clear central access to the interior volume.

The circular seating layout was selected after trade studies that examined linear, facing and radial arrangements. The radial configuration provides the best combination of structural symmetry, load distribution under multi-axis acceleration, and practical access for boarding, egress and in-flight communication among the crew.

Parameter	Value	Notes
External Diameter	3.60 m	Standard size class
External Height	2.80 m	Includes stowed leg envelope
Internal Usable Volume	≈ 18–20 m³	After structure and installed systems
Occupant Capacity	7 persons	Pilot + Co-pilot + 5 passengers
Seating Arrangement	Circular / Radial	Central access, improved G distribution
Target Operational Mass	7.2 – 7.8 tonnes	All systems + 7 occupants + full consumables
Structural Safety Factor	≥ 2.8	Primary cage under ultimate combined load
Centre-of-Gravity Target	Near geometric centre	Critical for stable descent and landing
Number of Openings	3 total	2 side doors + 1 roof hatch; continuous floor

2.1 Detailed Mass Budget

The following mass budget is the controlling target for all detailed design work. Any growth beyond the upper bound of 7.8 tonnes requires formal review and possible resizing of the parachute cluster and the landing-leg energy-absorption system. The budget includes a deliberate contingency margin to accommodate normal design evolution without immediate violation of the hard mass ceiling.

Group	Mass Range (kg)	Approx. Share
Primary structure & roll-cage	2 100 – 2 300	≈ 29 %
Landing legs + dampers + pads	850 – 950	≈ 12 %
Parachute system (canopies + mechanisms)	480 – 550	≈ 7 %
Separation actuators & interface hardware	320 – 380	≈ 5 %
Energy storage & electrical systems	650 – 750	≈ 9 %
Life support + 72 h consumables	420 – 480	≈ 6 %
Flotation, self-righting, location systems	280 – 340	≈ 4 %
Fire suppression, sealing, interiors	380 – 450	≈ 5 %
Seats, harnesses, human-factors equipment	350 – 400	≈ 5 %
Occupants (7 × 80 kg average)	560	≈ 7 %
Design margin / contingency	600 – 800	≈ 9 %
Total Operational Mass	7 200 – 7 800	100 %

3. Six Independent Safety Layers – Detailed Description

The defining feature of Version 1.3 is the expansion of the safety architecture to six fully independent layers. Each layer is capable of performing its critical function even if every preceding layer has failed. This eliminates single-point failures across the entire survival sequence: detection of the emergency condition, separation from the host, parachute deployment, controlled descent, impact attenuation, and post-landing survival and location. The layers are described in the order in which they are normally expected to act, but each remains independently capable.

Layer 1 – Primary Electrical Command and Sensing

Dual-redundant processors continuously monitor a suite of multi-axis accelerometers, rate gyroscopes, barometric and differential pressure sensors, and free-fall detection algorithms. The sensing suite is designed to recognise both rapid onset events (for example, mid-air collision or structural break-up) and more gradual but still critical conditions (for example, uncontrolled descent or prolonged high sink rate). Manual activation is provided by guarded push-buttons reachable from both the pilot and co-pilot positions. Power is drawn from the main emergency battery bus, which itself incorporates continuous health monitoring and automatic switch-over to backup sources. Under normal and most abnormal conditions Layer 1 is responsible for initiating host separation and parachute deployment. All critical commands are cross-checked between the two processors before actuation is authorised.

Layer 2 – Independent Pyrotechnic Path

A completely isolated pyrotechnic initiation architecture is powered by dedicated non-rechargeable thermal batteries. Full galvanic isolation from the main electrical system prevents common-mode electrical failures from disabling this path. Automatic arming occurs after a configurable timeout (typically three to five seconds) if Layer 1 has not confirmed successful separation or deployment. Layer 2 is capable of initiating both host-vehicle separation and parachute cluster deployment. The pyrotechnic devices are selected and qualified for the expected temperature, vibration, humidity and storage-life environment of the intended host platforms. Status of the pyrotechnic path is continuously monitored while the capsule is installed.

Layer 3 – Pure Mechanical Path (Zero Electronics)

High-force mechanical springs, steel cable runs and inertial masses provide a completely non-electrical and non-pyrotechnic activation path. A dedicated manual lever is positioned so that any of the seven occupants can reach it without requiring special tools or sequences. In addition, pure inertial triggering occurs automatically under high negative acceleration even in the absence of human action. Because this layer contains no electronics and no energetic materials beyond the stored spring energy, it remains functional after total electrical power loss, electromagnetic pulse events, or complete failure of the pyrotechnic system. It constitutes the ultimate non-energetic backup for separation and deployment initiation and is regarded as one of the highest-reliability elements of the entire architecture.

Layer 4 – High-Pressure Cold-Gas Deployment System

High-pressure bottles containing nitrogen or an equivalent inert gas are fitted with mechanical-thermal valves. If Layers 1–3 have not achieved confirmed parachute deployment, these valves open and the stored gas provides forced ejection and inflation assistance for the parachute cluster. The same gas system can supply supplementary force for host separation if required. Layer 4 is the final energetic deployment path and is designed to function after prolonged storage and under extreme temperature conditions. Bottle pressure and valve health are monitored while the capsule is installed in the host.

Layer 5 – Impact Energy Absorption and Structural Survival

Six mechanical landing legs are equipped with progressive hydraulic dampers and multi-rate elastomer pads. Viscoelastic elements in the seat suspensions provide an additional stage of energy absorption for the occupants. The continuous carbon-fiber floor and the full six-axis roll-cage (designed to a safety factor of at least 2.8) maintain a survivable internal volume under extreme multi-axis impact, roll-over and inversion loads. Layer 5 is the last line of protection if the entire parachute system fails to deploy or is destroyed. Its performance is verified by a combination of non-linear finite-element analysis and physical drop and impact testing at progressively higher energy levels, including off-axis and asymmetric cases.

Layer 6 – Post-Landing Survival, Flotation and Location

After a successful or partially successful landing, Layer 6 maximises the probability of timely rescue. Automatically deployed flotation volumes support the full operational mass with margin and, together with the external geometry, provide a self-righting capability after water impact or inversion. Dual-band emergency locators (406 MHz COSPAS-SARSAT beacon plus GNSS and two-way satellite messaging) together with high-intensity visual strobes and an acoustic beacon enable rapid location by search-and-rescue assets. A minimum 72-hour life-support package for seven occupants (oxygen, CO₂ scrubbing, water and basic rations) is carried. Dual-agent fire suppression remains available. The capsule is environmentally sealed against water and external contaminants. Manual activation and override controls for all Layer 6 functions are available to the occupants at all times.

4. Structural Architecture – Carbon-Fiber Roll-Cage

The primary load-carrying structure is a continuous six-axis carbon-fiber roll-cage constructed from large-section primary tubes (approximately 12 cm class) joined by precision titanium node fittings. The entire floor plane is continuous and contains no hatch or other discontinuity. This design decision maximises torsional and bending stiffness and provides a rigid, unbroken foundation for the six landing-leg attachment points.

Local reinforcement is provided around the two side-door apertures and the single roof hatch. Load paths from these openings are returned into the continuous cage network through the titanium nodes so that the global strength and stiffness of the cage are preserved. The target safety factor on primary members under ultimate combined loads (impact + roll + asymmetric landing) is ≥ 2.8 . Energy-absorbing interfaces between the cage and the seating system reduce the peak accelerations transmitted to the occupants. All external surfaces receive corrosion-resistant coatings suitable for long-term maritime and mixed-environment service.

The structural concept is intentionally simple and highly redundant. There is no reliance on single critical joints or on the integrity of any one door aperture. Even after significant local damage the remaining cage and continuous floor are expected to maintain a survivable volume for the occupants. This philosophy of graceful degradation under damage is central to the overall safety case.

5. Parachute System and Host Separation

5.1 Parachute Cluster

The recovery parachute system consists of six large primary canopies sized for controlled descent of the full operational mass, supplemented by eight auxiliary or stabiliser canopies that provide initial stabilisation and additional redundancy. Each primary canopy is fitted with dual pilot/extractor chutes whose function is to force opening of the main canopy if it is fouled or delayed. Inflation-confirmation sensors monitor each canopy; if confirmation is not received within a short, predefined window the next safety layer is automatically commanded. Both fully automatic and fully manual deployment modes are permanently available to the occupants. The system is designed so that the loss of one or more individual canopies still permits a survivable descent rate.

5.2 Host-Vehicle Separation System

Separation from the host is accomplished by parallel pyrotechnic and hydraulic release actuators. In addition, pure mechanical locks are provided that can be released by a manual lever even after total loss of electrical and hydraulic power. The separation plane geometry is arranged to minimise the risk of re-contact or entanglement with residual host structure. All electrical, data and fluid interfaces are designed for a clean break at the moment of separation. Pre-separation health monitoring and status indication are available on the host crew interface when the capsule is installed.

6. Impact Energy Management

Even in the worst-case scenario of complete parachute failure the capsule is required to give the occupants a high probability of survival. Energy management is achieved through a staged, progressive system that converts kinetic energy into controlled deformation and heat over a sufficiently long stroke.

1. The six mechanical legs deploy or extend to establish a stable landing attitude and to begin dissipating vertical kinetic energy.
2. Progressive hydraulic dampers absorb the majority of the remaining vertical energy over a controlled stroke.
3. Multi-rate elastomer pads provide the final stage of stroke and attenuate residual high-frequency shock.
4. Viscoelastic elements in the seat suspensions and the occupant restraint system further limit the peak loads experienced by the human body.
5. The continuous carbon-fiber floor and the roll-cage maintain a survivable internal volume under asymmetric, multi-axis and inversion loading.

The performance of the energy-absorption system is established by a combination of non-linear finite-element analysis and physical drop / impact testing at progressively higher energy levels. Off-axis and asymmetric impact cases are explicitly included in the test matrix because they represent realistic accident conditions.

7. Life Support, Environmental Control and Fire Protection

After landing or water impact the occupants may have to wait an extended period for rescue. The capsule therefore carries a minimum 72-hour life-support capability for seven persons. Oxygen is provided by a combination of stored gas and/or chemical generation, together with active CO₂ scrubbing and basic particulate and contaminant filtration. Emergency water and calorie-dense rations are sized for the same duration. Passive insulation supplemented by limited active thermal control maintains internal temperatures within survivable limits across the expected range of external conditions.

Fire protection employs a dual-agent architecture. A clean gaseous agent is used for the occupied interior volume so that the atmosphere remains breathable after discharge. A foam or powder capability is provided for external or residual fire threats. Coverage is designed to be omnidirectional. The capsule remains environmentally sealed after separation against water ingress and external contaminants. Fire detection is linked to the overall emergency logic so that an internal fire can, if necessary, contribute to the decision to separate from a burning host.

8. Flotation, Self-Righting and Emergency Location

For maritime and amphibious applications the capsule incorporates automatically deployed flotation volumes sized to support the full operational mass with an appropriate reserve of buoyancy. The external geometry, in combination with the flotation system, provides a self-righting capability that returns the capsule to an upright floating attitude after water impact or inversion. This is essential both for occupant comfort and for the proper operation of the location beacons, which are optimised for upright orientation.

Emergency location is provided by a dual-band system: a 406 MHz COSPAS-SARSAT beacon together with GNSS positioning and two-way satellite messaging. High-intensity visual strobes and an acoustic beacon further assist visual and auditory location by search-and-rescue teams. All location functions can be activated automatically upon separation or impact, or manually by the occupants at any time.

9. Door and Access Configuration

Access configuration was finalised in Version 1.2 and is retained without change in Version 1.3. The design balances structural strength against practical boarding and emergency egress requirements:

- **Two side doors** (port and starboard) serve as the primary emergency egress routes after separation and landing. They are sized and positioned for rapid exit by all seven occupants.

- **One large reinforced roof hatch** is the primary boarding path while the capsule is installed in the host vehicle and the primary external rescue access after landing.

- **No floor hatch** is provided. The entire bottom plane remains continuous. This decision maximises the strength and stiffness available to the six landing-leg attachment nodes and preserves the continuity of the roll-cage under the most severe impact and roll-over loads.

All apertures are locally reinforced. Load paths originating at the openings are returned into the primary cage structure through titanium nodes so that the global integrity of the roll-cage is not compromised.

10. Host Platform Interface Requirements

The capsule is conceived as a modular unit that can be integrated into multiple host platform types. Minimum interface requirements that any host must satisfy include:

- Structural hard-points capable of carrying the capsule mass under the full range of host crash, flight and operational loads, with appropriate safety factors.
- A defined separation plane incorporating controlled mechanical, electrical, data and fluid disconnects that achieve a clean break at the moment of separation.
- Provision for continuous pre-separation health monitoring and status display on the host crew interface.
- Sufficient clearance volume around the capsule to permit clean separation without interference from residual host structure.
- An optional umbilical for continuous power and data while the capsule is installed; the umbilical is automatically severed or disconnected at separation.

11. Verification and Validation Approach

A structured verification and validation programme is required before the design can be considered ready for certification or production release. The principal elements of that programme are outlined below. Each requirement in the specification will ultimately be linked to one or more methods of verification (analysis, inspection, demonstration or test).

- **Structural analysis and test** – Linear and non-linear finite-element analysis covering static, dynamic, impact and fatigue cases, correlated with physical static and dynamic testing of critical assemblies and of the complete cage.
- **Energy-absorption testing** – Full-scale or suitably scaled drop tests and impact tests of the landing-leg and damper system at progressively higher energy levels, including off-axis and asymmetric cases.
- **Parachute deployment trials** – Instrumented deployment tests under representative dynamic conditions, including deliberate failure-injection cases to verify the operation of successive safety layers.
- **Separation testing** – Controlled separation tests from representative host structures to confirm clean release, interface break and absence of re-contact.
- **Flotation and self-righting demonstrations** – Water-basin or open-water tests confirming flotation reserve and self-righting behaviour after inversion.
- **Life-support endurance** – Closed-capsule tests at full occupant load verifying atmosphere control, thermal environment and consumable endurance for the specified 72-hour period.
- **Environmental qualification** – Temperature, vibration, humidity, salt-fog and electromagnetic compatibility testing appropriate to the intended operational environments.
- **Human-factors evaluation** – Assessment of seating, harnesses, control reach, emergency procedures and egress under representative conditions.

12. Preliminary Patent-Family Assessment

The strengthened six-layer architecture, the continuous-floor high-strength carbon-fiber cage, the multi-path separation and deployment logic, the integrated progressive energy-absorption system, the flotation and self-righting capability, and the long-duration survival and location systems together support a broader and more defensible patent portfolio than the earlier four-layer baseline.

Estimated number of extractable patent families: 16 – 26

Of which judged high to very-high commercial and defensive strength: 10 – 16

Strongest conceptual clusters identified to date:

- Six-layer independent safety architecture specifically adapted to multi-domain survival modules.
- Continuous-floor carbon-fiber roll-cage combined with multi-leg progressive energy absorption.
- Multi-path (electrical / pyrotechnic / pure mechanical / cold-gas) deployment and separation logic.
- Integrated flotation, self-righting and dual-band location system enabling rapid multi-domain recovery.
- Modular host-interface and clean-break separation architecture suitable for diverse platform types.

13. Key Specification Summary

Item	Specification
Geometry	3.60 m diameter × 2.80 m height
Occupants	7 persons, circular / radial seating
Operational Mass	7.2 – 7.8 tonnes (controlling target)
Primary Structure	Carbon-fiber roll-cage, continuous floor, safety factor ≥ 2.8
Safety Layers	Six independent layers (Electrical, Pyrotechnic, Mechanical, Cold-Gas, Impact Absorption, Post-Landing Survival)
Parachute System	6 primary + 8 auxiliary canopies with multi-path deployment
Impact Absorption	6 mechanical legs + progressive hydraulic dampers + elastomer pads + viscoelastic seats
Life Support	≥ 72 hours for 7 occupants (O■, CO■ scrubbing, water, rations)
Fire Suppression	Dual-agent (clean gas interior + foam/powder exterior), omnidirectional
Flotation & Location	Automatic flotation + self-righting + dual-band 406 MHz / satellite beacons
Access Configuration	2 side doors + 1 reinforced roof hatch; fully continuous floor (no floor hatch)
Patent Potential	16–26 families (10–16 of high to very-high strength)

14. Document Control and Recommended Next Steps

This document (Document 00, Version 1.3 Full Detailed) constitutes the authoritative technical baseline for the Ultra-Safe Central Capsule. All subsequent detailed design, analysis, simulation, testing, certification planning and patent-extraction activities shall reference this baseline. Any material change to requirements, geometry, mass, safety-layer architecture or primary structural concept shall be captured in a formal revision of this document and shall trigger appropriate re-verification.

Recommended immediate next actions:

1. Freeze the external geometry and the controlling mass budget; release them for detailed CAD modelling.
2. Initiate high-fidelity finite-element models of the roll-cage, floor and leg-attachment regions.
3. Develop detailed logic diagrams, timing sequences and FMEA for Safety Layers 1 through 4.
4. Size the parachute cluster and the landing-leg stroke for the upper-bound operational mass of 7.8 tonnes.
5. Prepare host-interface control drawings for the priority platform applications.
6. Commission preliminary patent novelty searches on the strongest family clusters identified in Section 12.
7. Draft the top-level verification matrix that links each requirement to analysis, inspection or test methods.

— End of Document 00 (Version 1.3 Full Detailed) —

Ultra-Safe Central Capsule | Six Independent Safety Layers

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